

MRIIRS-Bank Management System

Detailed Implementation Steps

Step 1: Launch the Application. Open the terminal in the project directory and execute the compiled binary. The system greets the user with a clean, structured main menu.

```
blswa@myLab:~/Pictures/Bank management system$ ./bank_system

=====
          WELCOME TO MRIIRS BANK
=====
1. Create New Account
2. Deposit Money
3. Withdraw Money
4. Check Balance
5. Admin: List All Accounts
6. Exit
-----
Select Option: █
```

Step 2: Account Creation Select option 1 to register. The system automatically assigns the next available account number (e.g., 1002). Provide the full name, initial deposit, and a secure 4-digit PIN.

```
Select Option: 1

--- NEW ACCOUNT CONFIGURATION ---
YOUR ASSIGNED ACCOUNT NUMBER: 1002
Enter Full Name: Reehan Shazad
Initial Deposit Amount: 4000
Set Your 4-Digit Security PIN: 2000

Welcome to MRIIRS Bank, Reehan Shazad! Your account has been created.

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Select Option: █
```

Step 3: Depositing Funds To add money, select option 2. Authenticate using the account number and PIN, then enter the deposit amount. The system updates the balance instantly.

```
Select Option: 2

Enter Account Number: 1002
Enter PIN: 2000

--- WELCOME BACK, Reehan Shazad ---
Current Balance: 4000.00
Enter Amount to Deposit: 1000
Success! Your new balance is: 5000.00

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-----
Select Option: █
```

Step 4: Withdrawing Funds Select option 3 to withdraw. After authentication, enter the desired amount. The system validates the request against the current balance before processing.

```
Select Option: 3

Enter Account Number: 1002
Enter PIN: 2000

--- WELCOME BACK, Reehan Shazad ---
Current Balance: 5000.00
Enter Amount to Withdraw: 3000
Success! Your new balance is: 2000.00

=====
                WELCOME TO MRIIRS BANK
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6. Exit
-----
Select Option: █
```

Step 5: Checking Balance Quickly view your net balance by selecting option 4 and entering your credentials.

```
Select Option: 4

Enter Account Number: 1002
Enter PIN: 2000

--- WELCOME BACK, Reehan Shazad ---
Your Net Account Balance: 4000.00

=====
                WELCOME TO MRIIRS BANK
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6. Exit
-----
Select Option: █
```

Step 6: Administrative Overview Admin personnel can use option 5 to view a complete ledger of all accounts. This requires a secret master password for security.

```
Select Option: 5

Enter Admin Security Password: admin123

-----
Acc No      Account Holder      Balance
-----
1001        Test User                    1000.00
1002        Reehan Shazad                2000.00
1003        Abinash                      7000.00
1004        Sundeep B.                   20000.00
-----

=====
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-----
Select Option: █
```

Step 7: Graceful Exit Option 6 ensures all data is properly saved to the 'accounts.dat' file before closing the session.

```
=====
WELCOME TO MRIIRS BANK
=====
1. Create New Account
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3. Withdraw Money
4. Check Balance
5. Admin: List All Accounts
6. Exit
-----
Select Option: 6

Thank you for choosing MRIIRS Bank. Goodbye!
b1swa@myLab:~/Pictures/Bank management system$
```